

THREE HILLS, AB, Canada

WMO: 712490

Lat: 51.7675N

Lon: 113.2056W

Elev: 2812

StdP: 13.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-16.0	-27.4	1.3	-21.6	-21.3	1.9	-15.7	30.8	21.4	27.1	21.5	7.2	270	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	84.9	62.6	81.1	61.5	77.7	60.2	66.6	78.6	64.5	76.3	62.4	73.9	9.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	93.9	72.5	60.1	86.1	69.8	57.9	79.5	67.4	32.9	78.8	31.1	76.4	29.5	74.1	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	23.1	20.7	-28.1	90.6	6.3	4.0	-32.6	93.5	-36.3	95.8	-39.9	98.1	-44.4	101.0
			-28.3	71.3	6.3	2.8	-32.8	73.3	-36.5	74.9	-40.0	76.5	-44.6	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	13.6	16.2	25.5	39.1	50.2	57.3	62.2	60.7	52.5	39.9	26.1	15.2
	DBStd	20.74	15.82	14.09	14.34	9.57	8.18	5.61	5.43	6.13	8.26	9.21	12.94	14.55
	HDD50	5479	1128	948	759	342	98	7	1	4	66	329	718	1079
	HDD65	9811	1593	1368	1223	778	461	238	121	159	382	777	1167	1544
	CDD50	1215	0	0	0	13	105	226	378	335	140	17	1	0
	CDD65	73	0	0	0	0	2	7	33	25	6	0	0	0
	CDH74	1661	0	0	0	6	115	159	576	588	210	7	0	0
	CDH80	430	0	0	0	0	18	24	163	174	51	0	0	0
	WSAvg	9.7	9.7	9.3	9.6	10.9	11.5	10.6	8.5	8.2	9.3	9.8	9.4	9.3
	PrecAvg	16.0	0.7	0.6	0.7	0.9	1.8	2.8	2.8	1.9	1.8	0.7	0.6	0.6
Precipitation	PrecMax	22.7	2.2	1.3	1.9	2.4	4.4	6.7	6.1	4.9	5.0	1.8	1.8	1.6
	PrecMin	11.9	0.0	0.0	0.0	0.0	0.4	0.7	0.6	0.3	0.2	0.0	0.1	0.0
	PrecStd	2.8	0.6	0.4	0.5	0.6	1.0	1.4	1.5	1.3	1.2	0.5	0.4	0.3
	PrecStd	2.8	0.6	0.4	0.5	0.6	1.0	1.4	1.5	1.3	1.2	0.5	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	46.4	59.8	73.0	82.6	83.3	89.1	90.1	86.0	74.0	59.1	49.2
		MCWB	38.3	38.6	45.2	50.8	56.9	61.3	65.0	62.9	58.7	53.1	45.8	38.6
	2%	DB	41.3	41.3	52.0	66.9	77.2	78.5	84.6	84.7	80.0	65.5	52.2	43.1
		MCWB	36.0	35.4	41.3	48.6	54.3	59.5	65.8	61.8	57.4	49.7	42.1	36.6
	5%	DB	38.7	37.9	47.2	61.3	72.2	74.7	80.3	80.7	75.0	60.0	47.0	39.1
		MCWB	34.3	33.4	39.0	45.5	52.6	58.0	64.6	61.6	55.7	47.0	38.8	33.9
	10%	DB	34.9	34.2	42.4	55.8	67.5	71.2	76.3	76.5	69.4	55.1	42.2	35.0
		MCWB	31.5	30.7	36.6	42.7	50.5	56.6	62.8	60.7	53.9	44.6	36.0	31.0
	0.4%	WB	39.1	39.1	46.7	52.7	58.7	65.5	71.3	68.1	61.1	54.5	46.7	40.4
		MCDB	44.0	45.2	57.6	69.8	74.0	77.8	79.8	79.4	79.1	70.3	58.2	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	35.9	42.3	49.5	56.1	62.3	68.0	65.5	58.6	50.7	42.8	37.0
		MCDB	40.6	40.5	50.6	65.3	72.7	73.3	79.8	77.7	75.9	63.6	51.3	42.3
	5%	WB	34.5	33.5	39.5	46.4	53.8	60.0	65.8	63.5	56.5	47.9	39.5	34.0
		MCDB	38.5	37.5	46.4	59.1	69.3	71.1	77.7	76.1	72.3	58.5	46.1	38.9
	10%	WB	31.4	30.9	36.8	43.5	51.7	57.9	63.7	61.4	54.5	45.3	36.4	31.1
		MCDB	34.9	34.1	42.1	54.1	65.5	68.6	74.7	74.0	68.5	54.1	41.4	35.1
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.0	18.7	22.7	24.2	21.8	24.7	26.0	25.2	22.6	18.2	17.1
		MCDBR	19.2	20.6	25.0	33.0	32.6	29.0	31.5	34.6	35.3	30.7	24.2	20.3
		MCWBR	15.0	15.8	15.9	17.5	14.9	13.6	16.1	15.6	16.0	16.9	15.6	14.9
	5% WB	MCDBR	18.9	20.1	24.0	30.0	29.2	25.3	29.3	29.7	32.0	29.3	23.6	19.7
		MCWBR	15.3	15.7	16.1	16.6	13.7	13.4	16.5	15.4	15.3	16.6	15.7	14.8
Clear-Sky Solar Irradiance	taub		0.264	0.275	0.286	0.331	0.348	0.349	0.348	0.363	0.316	0.283	0.253	0.243
	taud		2.332	2.391	2.448	2.351	2.355	2.392	2.409	2.359	2.454	2.528	2.479	2.358
	Ebn at Noon		241	274	293	288	287	287	285	274	277	269	250	232
	Edh at Noon		19	24	28	35	37	36	35	35	28	21	16	16
All-Sky Solar Radiation	RadAvg		347	674	1113	1502	1822	1944	2065	1676	1175	707	373	268
	RadStd		27	56	92	124	119	118	103	118	124	83	32	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+508	+563	N/A	-34
Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page